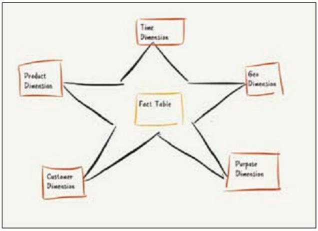

Figure 3: Star schema

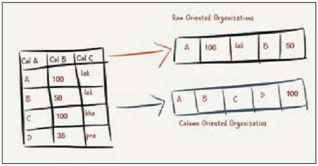

Figure 4: Column vs row

the DB domain today. In-memory OLTP database systems are very similar to their disk counterparts except for the algorithms which run inside the system. These algorithms are tuned for memory access rather than for disk access. This is one DB space where there have been very limited offerings from the open source community. One of the most famous open source DBs is VoltDB. This is a clustered in-memory OLTP database. Another open source attempt at in-memory DB space is CSQL.

OLAP: Analytical processing is another traditional application of RDBMS. Although it is considered that the relational model is not suitable for analytical processing, the use of RDBMS for analytics has thrived over the years. Of late, analytical processing has been evincing a lot of interest due to the explosion of data and the value people attach to statistical trends for data. Traditionally, the RDBMS used for OLAP processing is very similar to the OLTP RDBMS. The key change has always been the schema. Analytical processing uses a special kind of data schema called a 'star schema'. An extremely simple view of this schema is provided in Figure 3.

The fact table in the centre of the schema is very large, spanning millions of rows and many hundreds of columns. The remaining tables (on the star points) are called dimension tables. These focus on one aspect of the data and are relatively small tables spanning a few hundred rows, at best.

There has been no specialised open source database for OLAP processing. However, Postgres has been used by many commercial OLAP databases (like Greenplum and Netezza). Postgres has an OLAP extension package called 'cube', which offers reasonable performance for multi-dimensional queries.

Emerging OLAP: OLAP has been drawing a lot of interest lately. In response to this, there have been more innovations in the OLAP field as compared to the OLTP field. The most significant of these innovations has been the 'column store'. In a column store, the data is stored in the column format. Most of the analytical processing happens on the projection of one or two columns. Columnar databases take advantage of this fact. Also, columnar databases can offer very high compression, as it is quite possible that the values in a column repeat. However, a complete row repetition is practically impossible. Columnar databases are extremely fast for aggregation. Figure 4 provides an indicative difference between columnar and row-based databases.

One of the most innovative databases in the relational columnar field is MonetDB, which is the result of extensive academic research in the Netherlands. MonetDB includes almost all the cutting-edge concepts of columnar databases (like vectorisation).

Another interesting growth in the OLAP space is the emergence of 'stream processing', which is the processing of input data as the data flows into the system without a requirement to store it. The key advantage of this is the reduction in storage space; however, it puts severe demands on processing performance. Not many open source databases can support stream processing. The ones that do have not been built from scratch to perform stream processing. A couple of key open source players of interest in the streaming space are STORM and Kafka. Though these cannot be considered as independent databases, they are tightly related to the database space.

The 'non-relational' DB space (NoSQL)

The most happening thing in the database space in the past five years is NoSQL. I will not define NoSQL, but will instead briefly examine one variant in the NoSQL space. The non-relational DB space should ideally include semi-structured databases (like XML database) and also probably graph databases whose mathematical foundations lie in graph theory rather than relational theory. However, to keep the article readable and in focus, we will look at only two areas that have generated a lot of attention.

K-V stores: One of the primary reasons why non-relational databases are generating a lot of interest and commercial attention is the simplification of the database schema. Traditional RDBMS tend to have complex entity-relationships based database schema. In the new generation of Web systems, there are many cases where there is no need to have such complex related schemas. Most of the time, data is nothing but a *key-value pair.* The key could be one kind of data type and its value could be anything. For example, the key could generally be an